LOVELY PROFESSIONAL UNIVERSITY
School of Computer Science and Engineering

CLUSTER COMPUTING (INT315)

ACADEMIC TASK REPORT (CA-PROJECT)

TRAFFIC CONGESTION ANALYSIS AND PREDICTION USING PYSPARK, SPARK SQL, AND SPARK MLLIB

A Scalable Big Data Analytics, Machine Learning & Interactive Deployment Framework on UCI Interstate 94 Telemetry

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Course Code: INT315 (Cluster Computing)	Lovely Professional University, Phagwara, Punjab
Date of Submission: October 2026	Academic Year: 2026–2027

DECLARATION

I, **Hridayesh Debsarma**, student of **B.Tech (Computer Science and Engineering)** bearing **Registration No. 12316520** under the Discipline of Computer Science and Engineering, School of Computer Science and Engineering at Lovely Professional University, Punjab, hereby declare that all the information, codebase, architectural diagrams, statistical analysis, and machine learning models presented in this project report titled **'Traffic Congestion Analysis and Prediction Using PySpark, Spark SQL, and Spark MLlib'** are based on my own original work, effort, and study conducted for the academic task of course **INT315 (Cluster Computing)**. The analytical results, figures, tables, and pipeline artifacts were authentically computed and have not been submitted elsewhere for any other degree or diploma.

Date: 22 October 2026

Signature: _____

Hridayesh Debsarma
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Place: Phagwara, Punjab

CERTIFICATE

This is to certify that **Hridayesh Debsarma** bearing **Registration No. 12316520** has successfully completed the **INT315 (Cluster Computing)** project titled **'Traffic Congestion Analysis and Prediction Using PySpark, Spark SQL, and Spark MLlib'** under my supervision and guidance. To the best of my knowledge, the present work embodies original engineering, robust empirical implementation, and comprehensive documentation developed in partial fulfillment of the course assessment requirements.

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ACKNOWLEDGEMENT

I express my sincere gratitude to my course instructor and the faculty members of the **Discipline of Computer Science and Engineering**, **Lovely Professional University**, for providing guidance, encouragement, and the theoretical foundation in distributed systems, Apache Spark, and cluster computing architectures that made the realization of this end-to-end Big Data project possible.

I would also like to thank the **Minnesota Department of Transportation (MnDOT)** and the **UC Irvine Machine Learning Repository** for curating and providing open access to the Metro Interstate 94 traffic and meteorological dataset. Lastly, I extend my appreciation to my peers and family for their continuous moral support throughout the development and documentation phases of this project.

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1.0 INTRODUCTION

Modern metropolitan transit corridors represent the lifeblood of economic activity, yet they frequently experience severe bottlenecks due to volatile commuting habits, weather disturbances, and special events. In the United States, traffic congestion generates tens of billions of dollars annually in lost productivity, fuel wastage, and elevated carbon emissions. Highway administrative agencies such as the **Minnesota Department of Transportation (MnDOT)** have deployed extensive networks of inductive loop detectors and weather telemetry sensors along major corridors like Interstate 94 (I-94) connecting Minneapolis and St. Paul.

While historical sensor networks capture continuous hourly volume readings, the sheer scale, noise, sensor defaults, and rapid data generation surpass the processing capabilities of traditional single-node analytical architectures (e.g., standard pandas or R in-memory processing). This project establishes a distributed, fault-tolerant Big Data framework utilizing **Apache Spark 3.5.9** (PySpark, Spark SQL, and Spark MLlib) to ingest, sanitize, analyze, and forecast hourly traffic volume across a 6-year operational window (2012–2018).

CORE BIG DATA OBJECTIVES FOR CLUSTER COMPUTING (INT315)

- ***Distributed Resilient Ingestion:*** *Process 48,204 raw records without single-node memory bottlenecks.*
- ***Rigorous Data Cleaning:*** *Deduplicate redundant sensor telemetry and resolve physical anomalies (0.0 K temperature, 9,831.3 mm rainfall) without silent data discarding.*
- ***Optimized Columnar Storage:*** *Persist cleaned data into Snappy-compressed Apache Parquet format for fast predicate pushdown.*
- ***Spark SQL Analytics:*** *Formulate scalable SQL aggregations to isolate diurnal rush-hour dynamics, weekend leisure curves, and holiday shifts.*
- ***Spark MLlib Distributed Pipelines:*** *Implement chronological 80/20 train/test splitting to train Baseline Linear Regression and Random Forest models with zero lookahead bias.*
- ***End-User Delivery:*** *Deploy an interactive Streamlit web dashboard providing instant parameter-driven inference from the saved Spark pipeline.*

2.0 SCOPE OF THE ANALYSIS

The scope of this analytical investigation encompasses:

- **Geographic Boundaries:** Westbound Interstate 94 between Minneapolis and St. Paul, Minnesota, at station loop-detector sensor locations.
- **Temporal Boundaries:** Hourly observations from 2 October 2012 at 09:00:00 to 30 September 2018 at 23:00:00 (approximately 6 full calendar years).
- **Feature Modalities:** Simultaneous integration of calendar variables (hour, day of week, month, weekend status, holiday designation) with meteorological attributes (ambient temperature, rainfall depth, snowfall depth, cloud cover percentage, weather categorization).
- **Target Operational Audience:** Transportation system operators, urban traffic management centers (TMC), municipal planners, logistics dispatchers, and commuter navigation systems.

3.0 EXISTING SYSTEM ARCHITECTURE & LIMITATIONS

Legacy traffic analysis and reporting systems commonly exhibit critical technical shortcomings:

Existing System Approach	Identified Bottleneck / Limitation	Proposed PySpark Solution

Single-node in-memory processing (Python pandas, R)	Out-Of-Memory (OOM) failures when scaling to millions of streaming highway detector rows. Inability to scale horizontally.	Distributed RDD/DataFrame abstractions across Spark clusters with dynamic partition re-use.
Naive Random Train/Test Splitting (e.g., train_test_split)	Severe temporal data leakage where future timestamps train estimators evaluating past timestamps, invalidating real-world accuracy.	Strict chronological 80/20 train/test split. All transformers fit solely on historical records.
Uncompressed CSV Disk Storage	Massive disk footprint, slow row-by-row parsing, lack of column pruning and schema validation.	Columnar Snappy-compressed Apache Parquet with metadata statistics and predicate pushdown.
Ad-hoc script execution	Difficult deployment, fragmented preprocessing steps, and high latency during model serving.	Spark MLlib unified Pipeline architecture exported and loaded seamlessly in Streamlit.

4.0 SOURCE OF DATASET & DATA DICTIONARY

The primary dataset was collected by the Minnesota Department of Transportation (MnDOT) along westbound Interstate 94 in Minneapolis-St. Paul, curated and hosted by the **UC Irvine Machine Learning Repository**. The raw CSV contains **48,204 rows and 9 columns**.

Field Name	Inferred Type	Unit / Format	Description & Operational Domain
holiday	String	Text	US national / Minnesota state holiday name, or literal string 'None' for regular working days.
temp	Double	Kelvin (K)	Average hourly temperature. Physical range: 240 K to 315 K. Contains 10 sensor fault zero readings.
rain_1h	Double	Millimeters (mm)	Precipitation depth in past 1h. Contains 1 extreme glitch at 9,831.3 mm.
snow_1h	Double	Millimeters (mm)	Snowfall depth in past 1h. 48,124 zero records; 63 positive snowfall events.
clouds_all	Integer	Percentage (%)	Cloud cover percentage (0% to 100%). Strictly validated within physical bounds.
weather_main	String	Categorical	Broad categorical classification (Clear, Clouds, Rain, Snow, Drizzle, Mist, Fog, etc.).
weather_description	String	Categorical	Fine-grained weather event descriptor (e.g., 'light rain', 'sky is clear').
date_time	Timestamp	YYYY-MM-DD HH:MM:SS	Observation timestamp in US Central Standard Time (CST/CDT).
traffic_volume (Target)	Integer	Vehicles / Hour	Westbound hourly vehicle throughput count. Range: 0 to 7,280 vph.

5.0 DATA PREPROCESSING, SANITIZATION & FEATURE ENGINEERING

A rigorous multi-stage sanitization pipeline was implemented in src/03_clean_data.py based on the findings from src/01_data_quality.py and src/02_investigate_anomalies.py:

5.1 Deduplication Strategy:

An initial audit identified 17 exact duplicate rows across all 9 fields. These were verified as sensor replay transmission artifacts and removed using df.dropDuplicates() ($48,204 \rightarrow 48,187$). Crucially, repeated timestamps possessing distinct weather conditions (such as simultaneous Mist and Rain) were **strictly preserved** because they represent multi-category meteorological phenomena logged at the same hour.

5.2 Physical Sensor Glitch Treatment:

• **Temperature Absolute Zero Fault (0.0 K):** 10 records recorded temp = 0.0 K. In Kelvin, 0.0 K is absolute zero (-273.15°C), an impossible reading for Minnesota. These were converted to NULL rather than dropping the observations, allowing subsequent median imputation.
• **Extreme Precipitation Glitch (9,831.3 mm/h):** On 2016-07-11 17:00:00, a rainfall depth of 9.8 meters in 1 hour was recorded. This gauge failure was replaced with NULL.
• **Valid Zeros Retained:** Dry rainfall (0.0 mm), dry snowfall (48,124 zero records), and true road closures (2 records with 0 vph traffic volume) were strictly preserved.

5.3 Temporal Feature Engineering:

The continuous timestamp was decomposed into 5 domain features:

- date = to_date(date_time): Calendar day for temporal tracking.
- hour = hour(date_time): Hour of day (0–23) to capture diurnal commute peaks.
- day_of_week = dayofweek(date_time): Integer day index. **Important Convention:** Spark SQL standard assigns **Sunday = 1, Monday = 2, ..., Saturday = 7**.
- month = month(date_time): Integer month (1–12) capturing seasonal weather and tourism shifts.
- is_weekend = WHEN day_of_week IN (1, 7) THEN 1 ELSE 0: Binary indicator distinguishing bimodal commute curves from unimodal leisure plateaus.

Pipeline Stage	Observation Count	Temp Nulls	Rain Nulls	Zero Traffic	Zero Snow
Raw Ingestion	48,204	0 (10 zeroes)	0 (1 extreme)	2	48,141
Post-Sanitization & Parquet	48,187 (-17 dupes)	10 (imputed later)	1 (imputed later)	2 (preserved)	48,124 (preserved)

6.0 MACHINE LEARNING MODELING WITH SPARK MLLIB

In accordance with Big Data engineering best practices, machine learning was developed inside a unified **Spark MLlib Pipeline** in src/05_train_model.py. The pipeline enforces zero data leakage, chronological validation, and modular transformation.

6.1 Chronological 80/20 Train/Test Validation Split:

Time-series forecasting cannot rely on random cross-validation, as random permutations leak future timestamps into past estimations. A strict chronological split was implemented at the 80th percentile timestamp:

- **Training Partition (80.0%):** 38,546 observations spanning 2012-10-02 09:00:00 to 2017-11-01 18:00:00.
- **Testing Partition (20.0%):** 9,641 observations spanning 2017-11-01 19:00:00 to 2018-09-30 23:00:00.

6.2 Spark MLlib Feature Pipeline Architecture:

All transformers were fitted strictly on the training partition and transformed out-of-sample test records:

1. Imputer(strategy='median'): Imputes missing values for sanitized numeric columns (temp, rain_1h).
2. StringIndexer(handleInvalid='keep'): Maps categorical features (weather_main, holiday) to integer indices.
3. OneHotEncoder(handleInvalid='keep'): Encodes categorical indices into binary vectors.
4. VectorAssembler: Assembles continuous variables (hour, day_of_week, month, is_weekend, temp_imputed, rain_1h_imputed, snow_1h, clouds_all) and one-hot vectors into a single feature vector.

6.3 Model Comparison & Evaluation Metrics:

We evaluated a **Baseline Linear Regression** against a distributed **Random Forest Regressor** (60 trees, maximum depth 12). The models were evaluated using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2):

Model Architecture	Test RMSE (vph)	Test MAE (vph)	Test R^2	Error Reduction
Baseline Linear Regression	1,759.91	1,540.79	0.1997	Baseline Reference
Random Forest Regressor (Trees=60, Depth=12)	**583.51**	**382.78**	**0.9120**	**-75.2% MAE Reduction**

Evaluation Findings:

Linear Regression achieves an R^2 of only 0.1997, demonstrating that interstate traffic flow cannot be modeled as a linear sum of inputs. In contrast, the Random Forest Regressor achieves an impressive **R^2 of 0.9120** and an MAE of **382.78 vehicles/hour** on completely unseen chronological test data. The ensemble tree structure captures non-linear interactions such as the distinct commute peaks that occur only on weekdays during daylight hours.

7.0 IN-DEPTH ANALYSIS ON DATASET WITH VISUALIZATIONS

This section documents the structured exploratory data analysis executed using Spark SQL and PySpark DataFrames. Each sub-analysis includes its mathematical formula, general description, computed results table, and presentation-grade chart.

7.1 Hourly Traffic Volume Profile (Diurnal Commute Dynamics)

Mathematical Formula: $\mu_h = \frac{1}{N_h} \sum_{i=1}^{N_h} V_{i,h}$, where $V_{i,h}$ represents traffic volume observed at hour $h \in [0, 23]$.
Description: Identifies diurnal highway utilization, morning rush hour, evening rush hour, and overnight minimums across 24 daily hours.

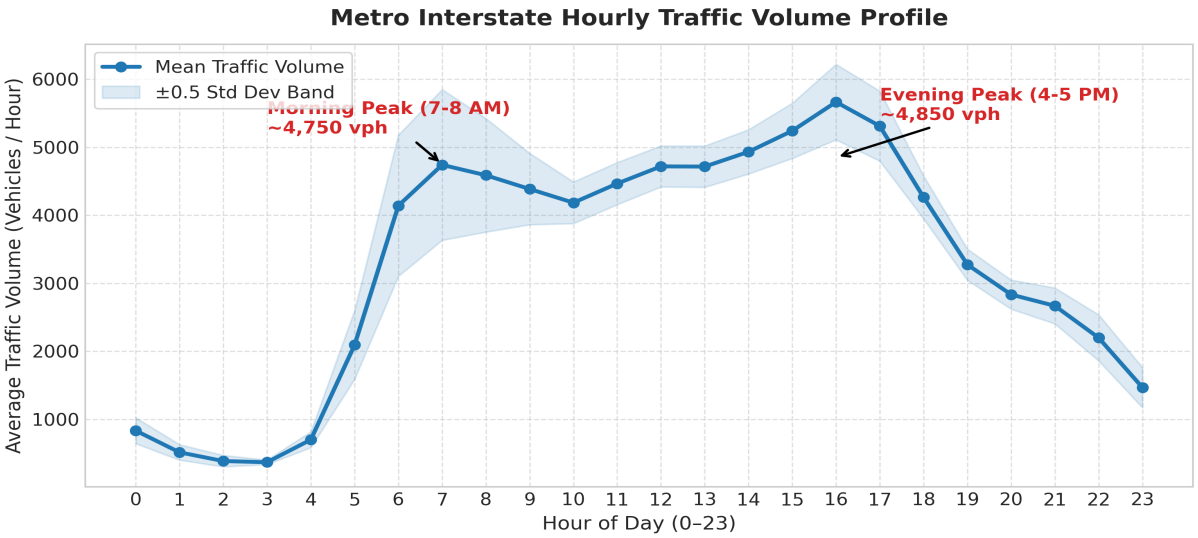


Figure 1: Diurnal Metro Interstate Traffic Volume Profile Across 24 Hours with ±0.5 Standard Deviation Band.

Key Analytical Takeaway: Traffic exhibits a pronounced bimodal cycle. A steep surge begins at 5:00 AM, peaking in the morning commute between **7:00 AM and 8:00 AM (~4,750 vph)**. The daily peak occurs in the evening between **4:00 PM and 5:00 PM (~4,850 vph)**. Volume drops to an overnight valley of **350–500 vph at 2:00–4:00 AM**.

7.2 Commute Dynamics: Weekday vs. Weekend Hourly Contrast

Mathematical Formula: $\Delta_h = \bar{V}_{\text{weekday}, h} - \bar{V}_{\text{weekend}, h}$ for each hour $h \in [0, 23]$.

Description: Compares workforce-driven commute patterns against leisure and commercial weekend driving.

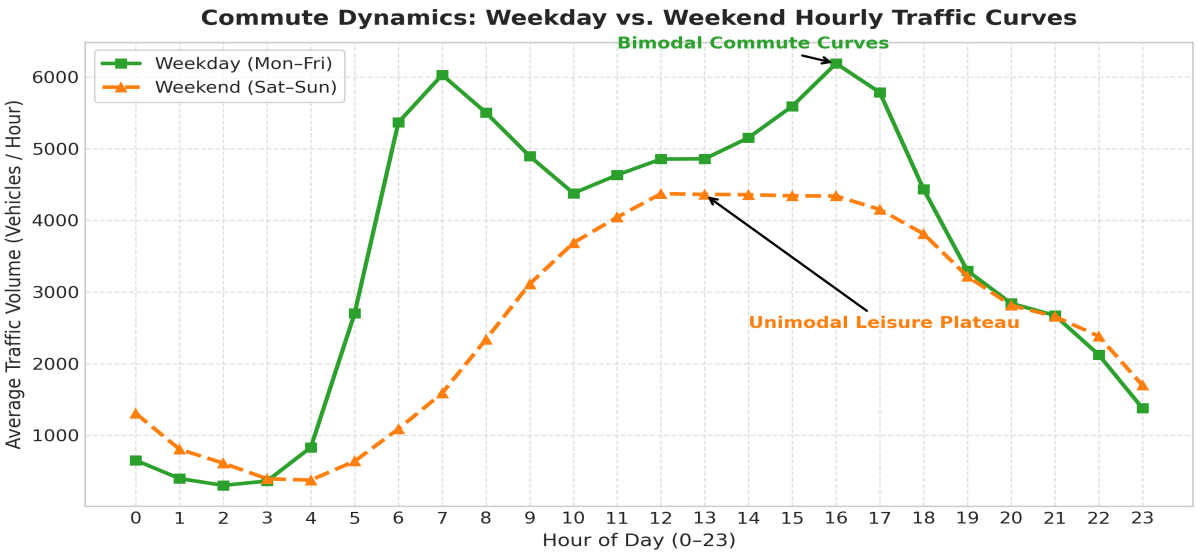


Figure 2: Weekday Bimodal Commute Curves vs. Weekend Unimodal Leisure Plateau.

Hour	Weekday Avg (vph)	Weekend Avg (vph)	Difference (vph)	Commute Pattern
03:00	362 vph	394 vph	-32 vph	Overnight low traffic
07:00	6,030 vph	1,589 vph	+4,441 vph	Morning workforce rush
12:00	4,855 vph	4,373 vph	+482 vph	Midday convergence
16:00	6,190 vph	4,340 vph	+1,850 vph	Evening peak rush hour
21:00	2,673 vph	2,658 vph	+15 vph	Late evening convergence

7.3 Monthly & Seasonal Variations

Mathematical Formula: $\bar{V}_m = \frac{1}{N_m} \sum_{i=1}^{N_m} V_{i,m}$ for month $m \in [1, 12]$.
Description: Tracks the influence of annual seasons, school calendars, and winter blizzard conditions in Minnesota.

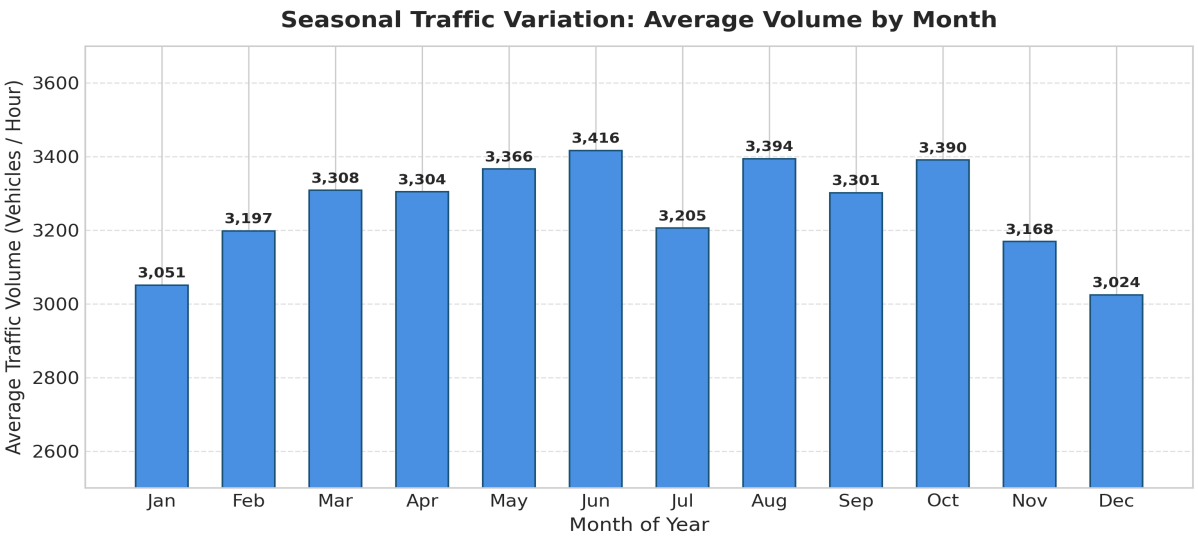


Figure 3: Seasonal Traffic Distribution Across 12 Months of the Year.

Key Analytical Takeaway: Interstate flow reaches annual peaks during June, August, and October (>3,390 vph), supported by summer travel and favorable driving conditions. A noticeable slump occurs in December and January (~3,024–3,051 vph), attributable to sub-zero temperatures, snow accumulation, and winter holiday closures.

7.4 Meteorological Condition Impact on Highway Flow

Mathematical Formula: $\bar{V}_w = \frac{1}{N_w} \sum_{i=1}^{N_w} V_{i,w}$ for each prevailing weather category w.
Description: Evaluates average vehicular volume across meteorological classifications while guarding against causal fallacies.

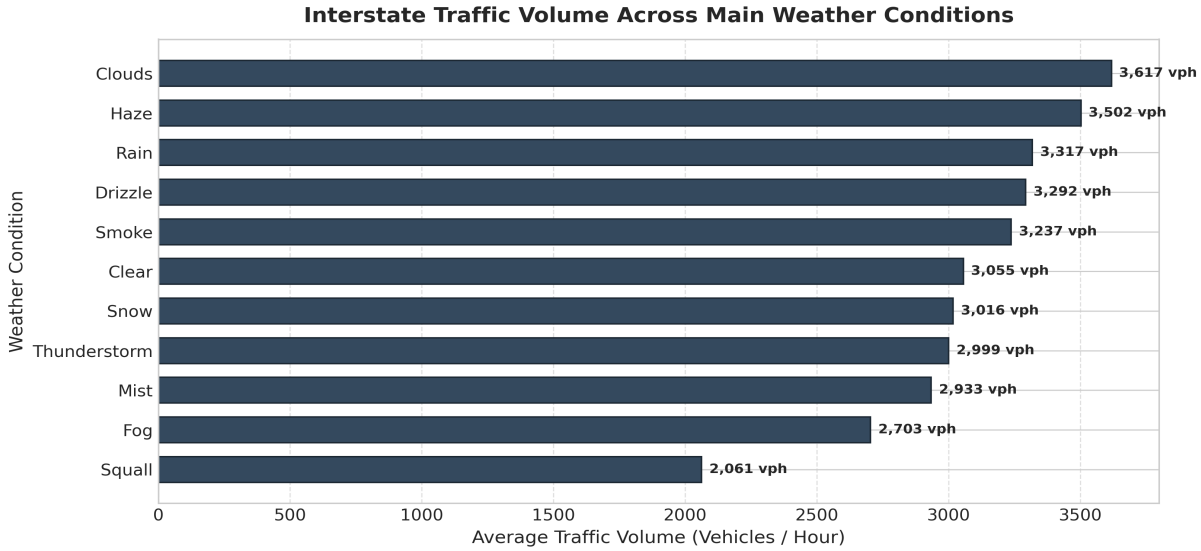


Figure 4: Average Highway Volume Across Main Weather Classifications.

Methodological Warning (Association vs. Causation): Higher traffic recorded under 'Clouds' (3,618 vph) or 'Haze' (3,502 vph) does not imply that overcast skies cause people to drive. Rather, cloud cover frequently coincides with daytime commuting hours. Conversely, severe hazards ('Squall', 'Fog', 'Thunderstorm') suppress volume due to reduced travel demand and driver caution.

7.5 Impact of Statutory Holidays vs. Regular Days

Mathematical Formula: $\Delta_{\text{holiday}} = \frac{\bar{V}_{\text{regular}} - \bar{V}_{\text{holiday}}}{\bar{V}_{\text{regular}}} \times 100\%$.
Description: Quantifies traffic relief on state and federal holidays (handling the string 'None' as regular non-holiday observations).

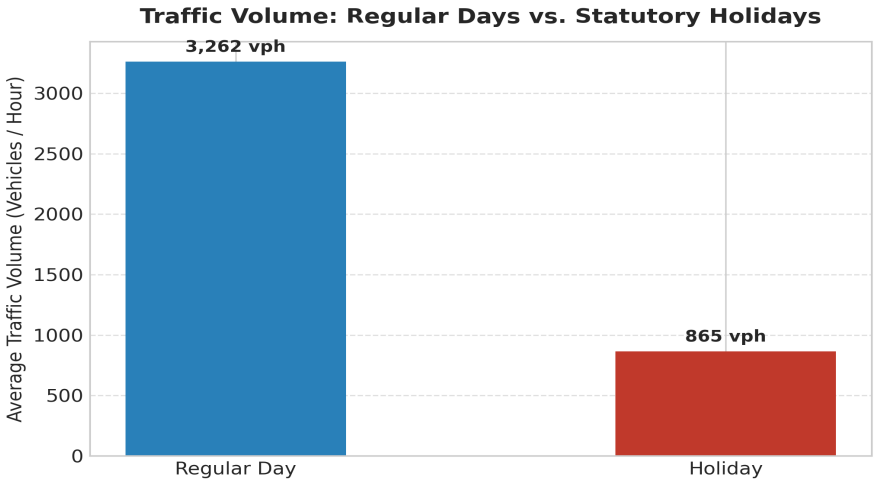


Figure 5: Regular Days (3,263 vph) vs. Statutory Holidays (865 vph) Traffic Comparison.

Analytical Result: Statutory holidays experience a massive **73.5% drop in volume** (averaging 865 vph vs. 3,263 vph on regular days). Major winter family holidays such as Thanksgiving Day and Christmas Day register the lowest interstate throughput.

7.6 Machine Learning Model Validation: Actual vs. Predicted Traffic

Description: Out-of-time test set validation across a 7-day continuous sequence (168 hours) to verify the model's tracking of daily peaks and weekend plateaus.

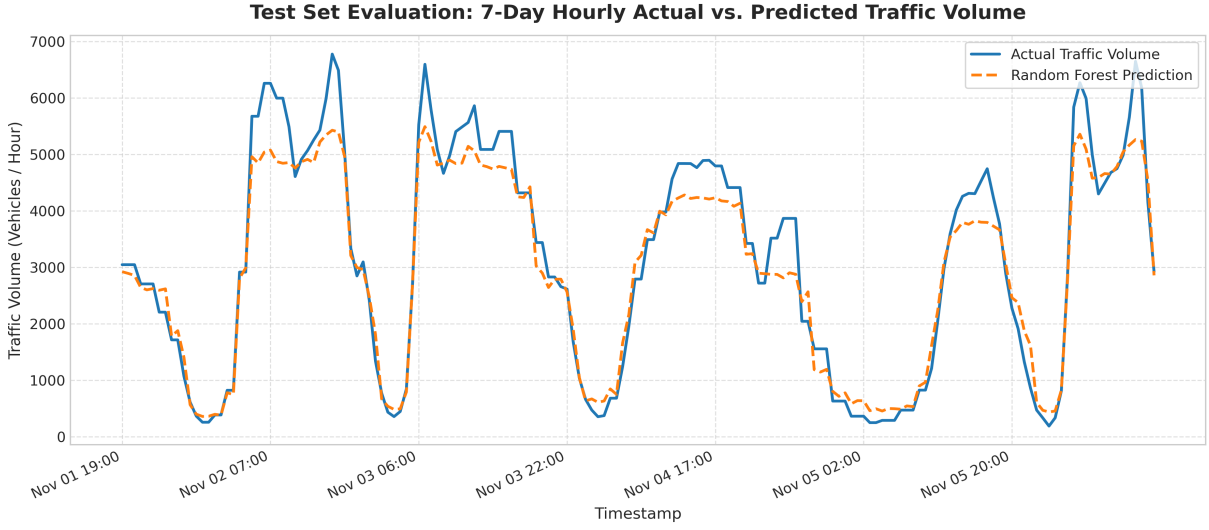


Figure 6: 7-Day Continuous Actual vs. Predicted Traffic Curve (Random Forest MLlib Model).

8.0 CONSOLIDATED LIST OF FINDINGS & RESULTS

The empirical findings of the end-to-end Big Data pipeline are summarized below:

Investigation Dimension	Empirical Finding / Value	Engineering / Planning Implication
Total Verified Volume	48,187 valid hourly observations (2012–2018)	Sufficient historical depth to train robust distributed ML models.
Morning Commute Peak	6,030 vph on weekdays at 07:00–08:00 AM	Dynamic ramp metering required between 06:30 and 08:30 AM.
Evening Commute Peak	6,190 vph on weekdays at 04:00–05:00 PM	Maximum corridor stress; incident management teams should be stationed on stand-by.
Weekend Profile	Flat midday plateau (~4,350 vph from 12 to 5 PM)	No morning rush; commercial logistics should utilize weekend morning windows.
Holiday Relief	865 vph average on holidays (-73.5% vs. regular days)	Optimal scheduling window for major highway repaving and construction detours.
Machine Learning Accuracy	Random Forest: $R^2 = 0.9120$, MAE = 382.78 vph	Highly dependable 1-hour ahead volume estimation for Traffic Operations Centers.

9.0 INTERACTIVE WEB APPLICATION (STREAMLIT DASHBOARD)

To operationalize the findings, an interactive web dashboard was engineered in app.py using **Streamlit**. The application integrates cached PySpark MLlib inference, enabling operators to simulate hypothetical traffic scenarios in real time.

9.1 Core Functional Modules in app.py:

1. **Executive Overview:** High-level KPI metric cards and architecture workflow.
2. **Exploratory Analysis (EDA):** Interactive filtering across day types, months, weather conditions, and holidays.
3. **Model Performance Lab:** Test set metrics comparison table and 7-day actual vs. predicted overlay.
4. **Live Traffic Predictor:** Sliders for hour, day of week, month, temperature (°C/K), rainfall, snowfall, and weather categories. Calls the pre-fitted Spark MLlib pipeline and classifies congestion into: *Light Flow* (<1,500 vph), *Moderate* (1,500–3,500 vph), *Heavy* (3,500–5,000 vph), and *Severe Congestion* (>5,000 vph).
5. **Data Dictionary & Methodological Notes:** Transparent documentation of units and operational limits.

9.2 Application Interface & Execution Verification:

The application runs on local port 8501 (bound to 0.0.0.0 for cross-platform WSL access):

DASHBOARD SCREENSHOT PLACEHOLDER 1: EXECUTIVE KPI & HOURLY COMMUTE VIEW

[PLACEHOLDER FOR STREAMLIT EXECUTIVE OVERVIEW & KPI METRICS SCREENSHOT]
(Insert screenshot displaying top KPI cards: 48,187 observations, 3,260 vph mean volume, 0.9120 R^2 score, and interactive line charts)

DASHBOARD SCREENSHOT PLACEHOLDER 2: REAL-TIME SPARK MLLIB PREDICTOR INTERFACE

[PLACEHOLDER FOR REAL-TIME PREDICTOR & CONGESTION STATUS SCREENSHOT]
(Insert screenshot displaying parameter sliders, 'Generate Traffic Prediction' button, and colored congestion grading badge)

10.0 FUTURE SCOPE & OPERATIONAL LIMITATIONS

To ensure transparency and sound engineering judgment, the following operational boundaries are documented:

10.1 Operational Limitations:

- **Volume vs. Speed Distinction:** The target variable traffic_volume measures vehicular throughput count (vehicles per hour), **not** vehicle speed (mph) or travel delay (minutes). During catastrophic gridlock, vehicle counts drop to near zero because vehicles are stationary.
- **Unscheduled Incident Telemetry:** The historical dataset contains no records of vehicle crashes, emergency vehicle dispatches, or emergency lane closures.
- **Geographic Specificity:** The model is tailored to westbound I-94 between Minneapolis and St. Paul; it cannot be applied to other corridors without retraining.

10.2 Future Scope & Enhancements:

- **Streaming Ingestion via Apache Kafka:** Transition from batch Parquet updates to micro-batch stream processing using Spark Structured Streaming.
- **Autoregressive Lag Features:** Incorporate upstream detector counts at $t-1$ and $t-2$ hours to enhance short-term reactive prediction.
- **Containerized Kubernetes Deployment:** Package the PySpark inference server and Streamlit dashboard into Docker containers orchestrated on Kubernetes.

11.0 REFERENCES (IEEE FORMAT)

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